GEOPHYSICAL RESEARCH LETTERS, VOL. 9, NO. 11, PAGES 1279-1282, NOVEMBER 1982

PALEOMAGNETIC DATA FROM THE COSO RANGE, CALIFORNIA AND CURRENT STATUS OF THE COBB MOUNTAIN NORMAL GEOMAGNETIC POLARITY EVENT

Edward A. Mankinen and C. Sherman Grommé

U.S. Geological Survey, Menlo Park, California 94025

Abstract. Two basalt flows which erupted about 1.08 m.y. ago in the Coso Range, California, have normal magnetic polarity and thus provide additional evidence for the Cobb Mountain normal polarity event. A review of available data confirms that this event was of geomagnetic origin. A mean age of 1.10 ± 0.02 m.y. B.P. for the Cobb Mountain normal polarity event was found to best fit all available radiometric and geologic data.

Introduction

Coso Hot Springs geothermal area is located in the Coso Range of southeastern California (Figure 1). Active fumaroles, hot springs, hydrothermally altered rock, and abundant young volcanic rock led to the selection of this area for study by the U.S. Geological Survey. Descriptions of the geology and geochronology may be found in Duffield et al. [1980] and Duffield and Bacon [1981].

Late Cenozoic volcanism in the Coso Range began about 6 m.y. ago [Bacon et al., 1982] with the largest volume of rock erupted between 4 and 2.5 m.y. ago [Duffield et al., 1980]. The volcanic rocks of this episode span the compositional range from basalt to rhyolite. A second period began about 1.1 m.y. ago and was strikingly bimodal in character [Bacon et al., 1980], with basalt generally preceding rhyolite of approximately the same age [Duffield et al., 1980]. The rhyolites of this episode are highly evolved magmas from a single voluminous silicic reservoir that apparently is the principal heat source for the present-day geothermal system [Bacon et al., 1980, 1981].

The first basalt flows related to the youngest phase of volcanic activity formed an eruptive group 1.08 ± 0.06 m.y. B.P. [Bacon, 1982] at the margin of what would later become the present rhyolite field. (No further eruption of basalt occurred until an episode at 0.412 ± 0.059 m.y. B.P.) Slightly later than but essentially contemporaneous with these basalts, a single rhyolite dome was erupted 1.04 ± 0.02 m.y. B.P. [Duffield et al., 1980]. Fluxgate magnetometer measurements during the course of geologic mapping indicated a normal polarity for the three basalt flows of this episode (W.A. Duffield, oral communication, 1978). Because these flows may have been erupted during the Cobb Mountain normal polarity event [Mankinen et al., 1978], samples for paleomagnetic study were obtained from the three basalt flows and the coeval rhyolite dome.

This paper is not subject to U.S. copyright. Published in 1982 by the American Geophysical Union.

Paper number 2L1549.

Paleomagnetism

Oriented cores were collected from the three approximately 1.08-m.y.-old basalt flows and the 1.04-m.y. rhyolite dome at the localities shown in Figure 1. The locality numbers are those used by Duffield et al. [1980]. Polarity and stability of natural remanent magnetization (NRM) in selected samples from each locality were tested by progressive alternating field (AF) demagnetization experiments. Samples from localities 35 and 58 were very stable and showed no directional change of NRM during pilot AF demagnetization, therefore magnetic cleaning of the remaining samples from these localities was not required. Samples from localities 32 and 33 were magnetically cleaned at peak fields of 50 and 150 oersteds, respectively, to remove all anomalous components of magnetization. Stable remanent magnetization directions for all the units are given in Table 1.

The rhyolite dome (loc. 58) proved to have a stable NRM and to have reversed polarity, as would be expected on the basis of its age. Initial measurements of the remanence directions of the three basalt flows indicated normal polarity, in accord with the results obtained in the field. Normal polarity at localities 32 and 35 was verified and the flows at these two localities were very likely erupted during the Cobb Mountain normal polarity event. Although samples from locality 33 originally yielded a mean NRM direction with an inclination of +70° and a declination of 18°E ($k = 48$, $\alpha_{95} = 8.1°$), this direction proved to be due to a normal polarity overprint of a primary reversed direction. The overprint is probably a viscous remanent magnetization (VRM) acquired during the Brunhes normal polarity epoch. The NRM's in samples from locality 33 were much weaker in original intensity than in those from the other localities and proved to be magnetically much less stable. The median demagnetizing field (MDF) for the NRMs at localities 32 and 35 was 100 and 200 oe, respectively, while the MDF at locality 33 was only 50 oe. Also, demagnetization to 100 oe reduced the NRM intensity of samples from locality 33 to one-tenth of its original value, whereas fields greater than 450 oe were necessary to reduce the intensity of samples from the other two localities by a similar amount. Additional experiments were carried out to determine if the differences in stability were mineralogically controlled.

Small cores (500 to 700 mg) were obtained from samples from each lava flow. Saturation remanence (J_{rs}) and saturation magnetization (J_s) of each minicore were measured, then the samples were saturated in a field of 9000 oe and thermomagnetic experiments were performed in an argon atmosphere using applied fields of 2500

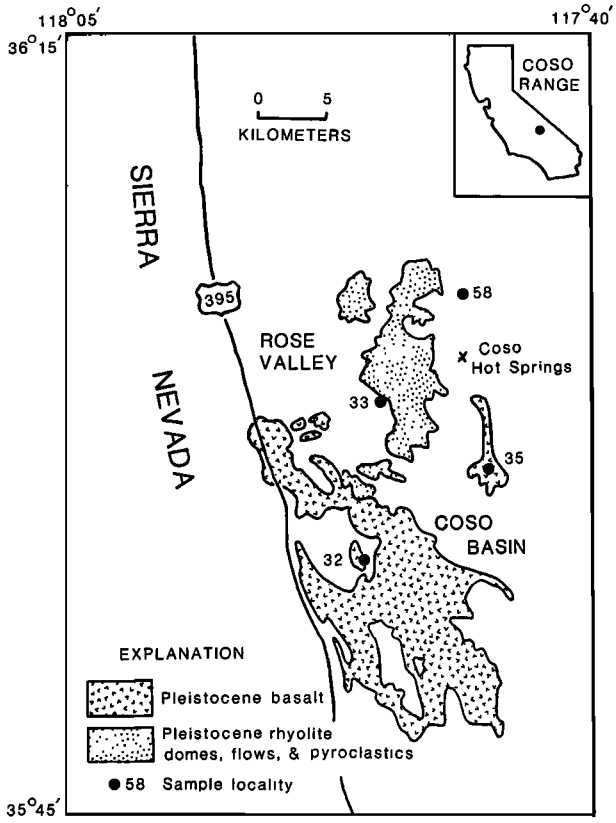


Figure 1. Sketch map of the Coso Hot Springs geothermal area showing main areas of Pleistocene volcanic rocks [after Duffield et al., 1980] and paleomagnetic sample localities.

and 3000 oe. The ratio J_{rs}/J_s ranges from 0.115 to 0.139 in these samples. The similarity of these ratios shows that there is very little difference in bulk magnetic hysteresis between the flows and that the magnetization is carried by large pseudo-single-domain to multidomain particles. Strong-field thermomagnetic experiments show similar behavior for the three flows, in all of which two Curie temperature distributions are present. The lower Curie temperatures are in the range of 150 to 200°C, the higher ones typically above 500°C. Thermomagnetic curves such as these are commonly observed in flows in which high-temperature deuteric oxidation has converted only part of an originally high-Ti titanomagnetite to a Ti-poor titanomagnetite containing ilmenite lamellae.

Weak-field susceptibility was found to range from 0.76 to 3.39 X 10^{-3} (c.g.s.) between these flows. However, this variation is probably due to differences in the volume fraction of magnetic minerals present, because the J_{rs}/J_s ratio indicates no large differences in magnetic grain size. Duffield et al., [1980] report large variations in opaque oxide content (up to 30%) in the Pleistocene basalt.

These experiments show that the three basalt flows are very similar in magnetic mineralogy and that another explanation is needed for the magnetic instability of samples from locality 33. One possibility is that this flow erupted very near in time to the polarity transition at either boundary of the Cobb Mountain normal polarity event. If this were true, a very weak remanence would likely have been acquired during a time when the magnitude of the earth's main dipole was greatly diminished and the flow would therefore be unusually susceptible to later overprinting. It has long been noted that sedimentary rocks deposited during polarity transitions have unstable magnetization; probably this is common in other rocks as well. One basalt flow, stratigraphically between normal and reversed polarity flows on Réunion Island in the Indian Ocean, is randomly magnetized and was thought by McDougall and Watkins [1973] to be unstable because of having been erupted during a polarity transition. Magnetic instability of the Huckleberry Ridge rhyolite tuff of Wyoming, which was probably erupted during the polarity transition at one of the boundaries of the Réunion event(s), was thought by Reynolds [1977] to be due in part to a weak initial magnetic intensity acquired when the earth's field was about one-tenth its normal intensity. In their study of the Tatoosh intrusion in Washington, Dodson et al. [1978] noted that the dispersion of directions increases markedly during the reversal of polarity. Two dacite flows from the Clear Lake, California volcanic field erupted at the end of the Jaramillo normal polarity event were magnetically very unstable [Mankinen et al.,

Table 1. Paleomagnetic results for volcanic rocks from the Coso Range.

Site	Calculated age* (m.y.)	N	H	Incl. (°)	Decl. (°)	J	k	α_{95}	PLA	PLO
58	1.04 ± 0.02	7	NRM	-42.5	184.6	0.99	214	4.1	-77.9	221.8
35	1.07 ± 0.14	8	NRM	+54.7	007.4	5.84	347	3.0	+83.9	336.9
32	1.07 ± 0.12	8	50	+53.5	358.8	8.16	94	5.7	+87.9	090.6
33	1.08 ± 0.06	8	150	-11.1	177.1	0.11	40	8.9	-59.5	247.8

N = number of samples; H = peak demagnetizing field (Oe); Incl. = inclination, positive downward; Decl. = declination, east of north; J = geometric mean intensity after magnetic cleaning, if needed (X 10^{-3} emu/cm^3); k = precision parameter (N-1/N-R); α_{95} = 95-percent confidence cone about average direction; PLA, PLO = north latitude and east longitude, respectively, of VGP position.

* from Duffield et al. [1980].

19448007, 1982, 11, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/GL009i011p01279 by Curtin University Library, Wiley Online Library on [25/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

1981], although many other similar units sampled in the same area behaved in a typical manner. Although the data from these studies are not conclusive, they do suggest the possibility of relative instability of NRM for flows erupted near or during polarity transitions.

Discussion

Paleomagnetic studies in the Clear Lake volcanic field, California, described anomalous magnetization directions in a rhyolite flow dated at 1.12 ±0.02 m.y. B.P. Apparent detection of the same event from other geographical areas suggested true geomagnetic field behavior and, therefore, the name "Cobb Mountain normal polarity event" was proposed [Mankinen et al., 1978]. Although two of the three basalt flows sampled in the Coso Range have age uncertainties slightly exceeding the minimum criteria of acceptability for defining the polarity time scale [Cox and Dalrymple, 1967], the eruptive episode to which all these flows belong clearly preceded the Jaramillo normal polarity event, which occurred between 0.97 and 0.90 m.y. B.P. [Mankinen et al., 1981]. The lavas must therefore have been erupted near to and during the time of the Cobb Mountain normal polarity event. The high virtual geomagnetic pole (VGP) positions calculated from the paleomagnetic data for the flows at localities 32 and 35 also indicate that a complete reversal of polarity occurred at this time.

Recently published studies have added new information concerning the nature of the Cobb Mountain normal polarity event and demonstrate that it was a global phenomenon. In their summary of magnetostratigraphic studies of the marine and loess deposits of the Ponto-Caspian region of the U.S.S.R., Kochegura and Zubakov [1978] showed a brief normal polarity interval immediately preceding the Jaramillo normal polarity event. This interval is referred to as the "Kvemonataneby excursion" and occurred approximately 1.1 m.y. B.P. No additional information concerning this interval is presented in their brief report.

A detailed record of a brief period of normal polarity preceding the Jaramillo normal polarity event in core KH73-4-7 from the Melanesian Basin was described by Kawai et al. [1977]. By correlating polarity boundaries in this core to known boundaries of polarity epochs and events, an estimate of 1.06 m.y. B.P. was obtained for the age of this normal episode [Sueishi et al., 1979].

Maenaka [1975) and Maenaka et al. [1977] reported normal polarity ash layers in the fluvial, lacustrine, and marine deposits of the Osaka Group, Japan that preceded the Jaramillo normal polarity event. One ash layer yielded a fission-track age of 1.1 ± 0.1 m.y. B.P. [Nishimura and Sasajima, 1970]. Maenaka [1975] believed that this was a reasonable age based on geologic arguments and suggested that a new polarity event might be represented by these data. More than 120 oriented cores spanning a thickness of 26 meters have now been obtained from the ash horizons and adjacent sediments [Maenaka, 1979]. A long sequence of normal polarity was found, followed in turn by a transitional polarity zone and a reversed polarity zone, which led Maenaka [1979] to tentatively name the normal polarity zone as the "Komyoike event" after the type locality of the Komyoike volcanic ash layer.

In their study of the geochronology and paleomagnetism of volcanic rocks from the Lesser Antilles, Briden et al. [1979] reported an intermediate polarity lava flow at 1.13 m.y. B.P. and six normal polarity flows ranging in age from 1.07 to 1.37 m.y. B.P. On the basis of these data, the authors suggested the possibility of a normal polarity event at about 1.18 m.y. B.P. However, this age is an unweighted mean of the seven lava flows, one of which has an unacceptably large uncertainty (1.37 ± 0.20 m.y.). If this determination is discarded, the weighted mean age of the remaining lava flows is 1.13 ± 0.02 m.y. B.P., which is in excellent agreement with the age of 1.12 ± 0.02 m.y. B.P. reported for the Cobb Mountain normal polarity event [Mankinen et al., 1978].

Combining information presented in this discussion with data previously considered likely to represent the Cobb Mountain normal polarity event [Mankinen et al., 1978], there can be little doubt that a geomagnetic event of global significance occurred just prior to the Jaramillo normal polarity event. Data from the Coso Range, Japan, and the Lesser Antilles also show that this event was a complete reversal of polarity rather than an excursion of the VGP to low latitudes. Mankinen and Dalrymple [1979] listed five lava flows with normal or intermediate polarity that are significantly older than the Jaramillo normal polarity event with ages ranging from 1.03 to 1.25 m.y. B.P. Although the occurrence of more than one event during this period of time cannot be completely ruled out, there is currently no reason to suppose that these data do not define a single event. Combining these ages with those from Cobb Mountain and the Lesser Antilles yields a weighted mean age of 1.10 ± 0.02 m.y. B.P. We believe that the mean age of 1.10 ± 0.02 m.y. B.P. best fits all the available radiometric and paleomagnetic data and is therefore a reasonable estimate of the true age of the Cobb Mountain normal polarity event.

Because the Cobb Mountain normal polarity event has rarely been detected in deep-sea cores and marine magnetic anomalies, it was believed to have lasted about 10,000 years or less [Mankinen et al., 1978]. A duration of 8,500 years determined by Sueishi et al. [1979] for the same event is in excellent agreement with the earlier estimate. A compilation of polarity and age data by Mankinen et al. [1981] shows 16 reversely magnetized lava flows within the age range of those having normal polarity as discussed above. The number of both normal and reversed polarity lava flows with analytically indistinguishable ages is further evidence that the event must have been very brief. Depending on the actual number of brief events within the Brunhes normal polarity epoch, Champion et al. [1981] estimated their average length to be between 4,000 and 10,000 years. A length of 10,000 years would therefore appear to be a

19448007, 1982, 11, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/GL009i011p01279 by Curtin University Library, Wiley Online Library on [25/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

reasonable estimate of the maximum duration of such brief events. Precise estimates are difficult to obtain because such short lengths of time are beyond the resolution of radiometric dating techniques.

Despite the fact that the Cobb Mountain normal polarity event may have lasted less than 10,000 years, there is some indication that the length of time was sufficient for the geomagnetic field to attain an approximately normal intensity. The flows at sites 32 and 35 have NRM intensities of 1.16×10^{-2} and 5.84×10^{-3} emu/cm^3 respectively. The in situ Konigsberger ratio, Q, is 11 for the flow at site 32 and 15 for that at site 35. Both values are well within ranges typically found for basalt flows that were not erupted during polarity transitions. The initial NRM intensity and corresponding Q ratio could not be determined for the flow at site 33 because of the normal polarity overprint. However, the geomagnetic field intensity at the time when the flows at sites 32 and 35 were erupted must have been stronger than when the flow at site 33 was erupted if our explanation of the unstable magnetization at site 33 is correct. Neither NRM intensities nor Q values were published for the other flows that may record the Cobb Mountain normal polarity event, so it is not known if normal geomagnetic field intensities were recorded by these flows as well.

Acknowledgments. We thank personnel of the China Lake Naval Weapons Center for access to the field area and C.R. Bacon for aiding in the collection of samples. We also thank D. E. Champion and R. L. Reynolds for helpful reviews.

References

Bacon, C. R., Time-predictable bimodal volcanism in the Coso Range, California, Geology, 10, 65-69, 1982.

Bacon, C. R., W. A. Duffield, and K. Nakamura, Distribution of Quaternary rhyolite domes of the Coso Range, California: Implications for extent of the geothermal anomaly, J. Geophys. Res., 85, 2525-2433, 1980.

Bacon, C. R., D. M. Giovannetti, W. A. Duffield, G. B. Dalrymple, and R. E. Drake, Age of the Coso Formation, Inyo County, California, U.S. Geol. Surv. Bull., in press, 1982.

Bacon, C. R., R. MacDonald, R. L. Smith, and P. A. Baedecker, Pleistocene high-silica rhyolites of the Coso volcanic field, Inyo County, California, J. Geophys. Res., 86, 10223-10241, 1981.

Briden, J. C., D. C. Rex, A. M. Faller, and J. F. Tomblin, K-Ar geochronology and palaeomagnetism of volcanic rocks in the Lesser Antilles arc, Philosophical Trans. R. Soc., London, 291A, 485-528, 1979.

Champion, D. E., G. B. Dalrymple, and M. A. Kuntz, Radiometric and paleomagnetic evidence for the Emperor reversed polarity event at 0.46 ±0.05 m.y. in basalt lava flows from the eastern Snake River plain, Idaho, Geophys. Res. Lett., 8, 1055-1058, 1981.

Cox, A., and G. B. Dalrymple, Statistical analysis of geomagnetic reversal data and the precision of potassium-argon dating, J. Geophys. Res., 72, 2603-2614, 1967.

Dodson, R., J. R. Dunn, M. Fuller, I. Williams, H. Ito, V. A. Schmidt, and Y-M. Wu, Paleomagnetic record of a late Tertiary field reversal, Geophys. J. R. Astron. Soc., 53, 373-412, 1978.

Duffield, W. A., and C. R. Bacon, Geologic map of the Coso volcanic field and adjacent areas, Inyo County, California, scale 1:50,000, miscellaneous geological investigation map, I-1200, U.S.. Geol. Surv., Menlo Park, Calif., 1981.

Duffield, W. A., C. R. Bacon, and G. B. Dalrymple, Late Cenozoic volcanism, geochronology, and structure of the Coso Range, Inyo County, California, J. Geophys. Res., 85, 2381-2404, 1980.

Kawai, N., T. Sato, T. Sueishi, and K. Kobayashi, Paleomagnetic study of deep-sea sediments from the Melanesian basin, J. Geomag. Geoelectr., 29, 211-223, 1977.

Kochegura, V. V., and V. A. Zubakov, Paleomagnetic time scale of the Ponto-Caspian Plio-Pleistocene deposits, Palaeogeography, Paleoclimatology, Palaeoecology, 23, 151-160, 1978.

Maenaka, K., The polarity change obtained from the water-laid volcanic ash layers in Plio-Pleistocene sediments in Kinki and Tokai district, Rock Magnetism and Paleogeophysics, 3, 41-46, 1975.

Maenaka, K., Paleomagnetic study of sediments around the Komyoike volcanic ash horizon in Osaka Group, Senpoku area, Osaka Prefecture, Japan, Geophys. Res. Lett., 6, 257-260, 1979.

Maenaka, K., T. Yokoyama, and S. Ishida, Paleomagnetic stratigraphy and biostratigraphy of the Plio-Pleistocene in the Kinki district, Japan, Quaternary Res., 7, 341-362, 1977.

Mankinen, E. A., and G. B. Dalrymple, Revised geomagnetic polarity time scale for the interval 0-5 m.y. B.P., J. Geophys. Res., 84, 615-626, 1979.

Mankinen, E. A., J. M. Donnelly, and C. S. Grommé, Geomagnetic polarity event recorded at 1.1 m.y. B.P. on Cobb Mountain, Clear Lake volcanic field, California, Geology, 6, 653-656, 1978.

Mankinen, E. A., J. M. Donnelly-Nolan, C. S. Grommé, and B. C. Hearn, Jr., Paleomagnetism of the Clear Lake volcanics and new limits on the age of the Jaramillo normal-polarity event, in Research in the Geysers-Clear Lake Geothermal Area, Northern California, ed. by R. J. McLaughlin and J. M. Donnelly-Nolan, p. 67-82, U.S. Geol. Surv. Prof. Paper 1141, 1981.

McDougall, I., and N. D. Watkins, Age and duration of the Réunion geomagnetic polarity event, Earth Planet. Sci. Lett., 19, 443-452, 1973.

Nishimura, S., and S. Sasajima, Fission track age of volcanic ash layers of the Plio-Pleistocene series in Kinki district, Japan (in Japanese with English abstract), Chikyu Kagaku (Earth Science), 24, 222-224, 1970.

Reynolds, R. L., Paleomagnetism of welded tuffs of the Yellowstone Group, J. Geophys. Res., 82, 3677-3693, 1977.

Sueishi, T., T. Sato, N. Kawai, and K. Kobayashi, Short geomagnetic episodes in the Matuyama epoch, Physics Earth Planet. Inter., 19, 1-11, 1979.

(Received August 24, 1982;
accepted September 27, 1982.)

19448007, 1982, 11, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/GL009i011p01279 by Curtin University Library, Wiley Online Library on [25/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License